

Fraud Detection & Transaction Risk Analysis

SQL Project Documentation | SQLite | 1.2M+ Transactions

Dataset Used: Credit Card Fraud Detection Dataset from Kaggle

Query 1 — Fraud Overview

SQL Query

```
SELECT
    COUNT(*) AS total_transactions,
    SUM(is_fraud) AS total_fraud_cases,
    ROUND(100.0 * SUM(is_fraud) / COUNT(*), 2) AS fraud_rate_pct,
    ROUND(SUM(amt), 2) AS total_amount_processed
FROM transactions;
```

Result Table:

<u>total_transactions</u>	<u>total_fraud_cases</u>	<u>fraud_rate_pct</u>	<u>total_amount_processed</u>
1,296,675	7,506	0.58%	\$91,222,428.90

Key Insights

- Over **1.29 million credit card transactions** were analyzed.
- **7,506 transactions were identified as fraudulent**, which is about **0.58%** of all transactions.
- The total value of all processed transactions was over **\$91.2 million**.
- Even though fraud is relatively rare, its presence calls for careful monitoring and anomaly detection to catch suspicious activity early.

Query 2 — Fraud-Prone Merchant Categories

SQL Query

```
SELECT category,
    COUNT(*) AS total_transactions,
    SUM(is_fraud) AS fraud_cases,
    ROUND(100.0 * SUM(is_fraud) / COUNT(*), 2) AS fraud_rate_pct,
    ROUND(SUM(amt), 2) AS total_fraudulent_amount
FROM transactions
GROUP BY category
ORDER BY fraud_rate_pct DESC
LIMIT 5;
```

Result Table:

<u>category</u>	<u>total transactions</u>	<u>fraud cases</u>	<u>fraud rate pct</u>	<u>total fraudulent amount</u>
shopping_net	97,543	1,713	1.76%	\$8,625,149.68
misc_net	63,287	915	1.45%	\$5,117,709.26
grocery_pos	123,638	1,743	1.41%	\$14,460,822.38
shopping_pos	116,672	843	0.72%	\$9,307,993.61
gas_transport	131,659	618	0.47%	\$8,351,732.29

Key Insights

- Online shopping (referred to as *shopping_net*) had the highest fraud rate at **1.76%**, showing that online retail is a big target for fraud.
- Grocery stores processed the highest number of fraud cases (**1,743 cases**) and the largest amount of fraudulent money, exceeding **\$14.4 million**.
- Fraud tends to happen more in consumer facing categories, especially where transactions are digital or card-not-present, such as online shopping and miscellaneous network transactions.
- Traditional categories like gas and transport had lower fraud rates.

Query 3 — High-Risk Transaction Classification

SQL Query

```
SELECT merchant, category, amt, city, state,  
CASE  
    WHEN amt > 500 AND is_fraud = 1 THEN 'Confirmed High Risk'  
    WHEN amt > 500 AND is_fraud = 0 THEN 'Flagged for Review'  
    WHEN amt BETWEEN 100 AND 500 AND is_fraud = 1 THEN 'Medium Risk Fraud'  
    ELSE 'Low Risk'  
END AS risk_classification  
FROM transactions  
WHERE amt > 100  
ORDER BY amt DESC  
LIMIT 50;
```

Result Table:

<u>merchant</u>	<u>category</u>	<u>amt</u>	<u>city</u>	<u>state</u>	<u>risk_classification</u>
fraud_Satterfield-Lowe	travel	\$28,948.90	Westerville	NE	Flagged for Review

fraud_Monahan, Hermann and Johns	travel	\$27,390.12	Armonk	NY	Flagged for Review
fraud_Monahan, Hermann and Johns	travel	\$27,119.77	Amorita	OK	Flagged for Review
fraud_Boyer-Haley	travel	\$26,544.12	Cross	SC	Flagged for Review
fraud_Hackett Group	travel	\$25,086.94	Lawrence	MA	Flagged for Review

Top high-risk transactions shown above. Full analysis identified recurring fraud patterns in online shopping and high-value merchant categories.

Key Insights

- Transactions over **\$10,000**, especially in the travel category, were flagged as high risk.
- A classification system was used to label transactions as “Confirmed High Risk,” “Flagged for Review,” or “Medium Risk Fraud” based on amount and fraud status.
- These flagged transactions were spread across multiple US states, showing that fraud isn’t isolated to one region.
- Large amounts combined with specific merchant categories can help detect potential fraud faster.

Query 4 — Geographic Fraud Analysis

SQL Query

```
SELECT state,
       COUNT(*) AS total_transactions,
       SUM(is_fraud) AS fraud_cases,
       ROUND(AVG(amt), 2) AS avg_transaction_amount,
       ROUND(100.0 * SUM(is_fraud) / COUNT(*), 2) AS fraud_rate_pct
FROM transactions
GROUP BY state
ORDER BY fraud_cases DESC
LIMIT 10;
```

Result Table:

<u>state</u>	<u>total transactions</u>	<u>fraud cases</u>	<u>avg transaction amount</u>	<u>fraud rate pct</u>
NY	83,501	555	\$71.93	0.66%
TX	94,876	479	\$71.68	0.50%

PA	79,847	458	\$72.28	0.57%
CA	56,360	326	\$73.42	0.58%
OH	46,480	321	\$73.07	0.69%
FL	42,671	281	\$73.94	0.66%
IL	43,252	248	\$69.64	0.57%
MI	46,154	238	\$71.13	0.52%
AL	40,989	215	\$65.45	0.52%
MN	31,714	207	\$69.43	0.65%

Key Insights

- New York had the highest number of fraud cases (**555 cases**), reflecting its high volume of transactions.
- Ohio had the highest fraud rate at **0.69%** despite fewer transactions, indicating a concentrated fraud problem.
- Average transaction amounts were fairly consistent across states, usually between **\$65** and **\$74**.
- Fraudulent activity was spread across many states, not just focused in one area, so monitoring needs to be nationwide.

Query 5 — Spending Anomaly Detection

SQL Query

```
SELECT category,
       ROUND(AVG(CASE WHEN is_fraud = 1 THEN amt END), 2) AS avg_fraud_amount,
       ROUND(AVG(CASE WHEN is_fraud = 0 THEN amt END), 2) AS avg_legitimate_amount,
       ROUND(
         AVG(CASE WHEN is_fraud = 1 THEN amt END) -
         AVG(CASE WHEN is_fraud = 0 THEN amt END),
         2) AS spend_anomaly_gap
FROM transactions
GROUP BY category
ORDER BY spend_anomaly_gap DESC;
```

Result Table:

<u>category</u>	<u>avg_fraud_amount</u>	<u>avg_legitimate_amount</u>	<u>spend_anomaly_gap</u>
shopping_net	\$999.25	\$72.14	\$927.11
shopping_pos	\$876.92	\$73.98	\$802.94
misc_net	\$797.01	\$70.36	\$726.65
entertainment	\$503.54	\$63.12	\$440.42
home	\$257.43	\$57.95	\$199.48
grocery_pos	\$311.99	\$114.17	\$197.82
misc_pos	\$218.28	\$62.40	\$155.88

Key Insights

- Fraudulent transactions generally involved much higher amounts than legitimate ones across all merchant categories.
- Online shopping had the biggest gap between fraud and legitimate spending. On average, fraudulent transactions were about **\$927** higher.
- Digital and card-not-present transactions were more vulnerable to fraud compared to traditional retail.
- The amount spent is a strong indicator of risk and can help improve fraud detection systems.

Final Project Summary

This project used SQL queries in SQLite to analyze over **1.29 million** credit card transactions. It uncovered key fraud patterns, including:

- Total fraudulent transactions: **7,506**
- Fraud rate: **0.58%**
- Total transaction value analyzed: **\$91.2 million+**

Key takeaways:

- Online shopping is the riskiest category for fraud.
- High value travel transactions often trigger fraud alerts.
- Fraud occurs across many US states, not just in isolated regions.
- Fraudulent transactions tend to involve much larger amounts than legitimate ones.

SQL techniques utilized include aggregation, grouping, conditional CASE statements, and classification rules to identify and understand fraud risks.